

EcoAlert - LoRa Based Smart Wildlife Detection And Alert System

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Abstract—Animal-train collisions present a significant challenge in regions where animals cross railway tracks. Such accidents result in loss of life and damage to railway infrastructure. The proposed system employs a MobileNet V2 CNN-based early warning system to mitigate these incidents. The system utilizes CNN-based image processing, along with an IR sensor and a camera, to detect animals. Multiple detection units are deployed along railway tracks. The prototype system processes data using a laptop as a substitute for a Raspberry Pi, with a NodeMCU controlling components such as GSM, GPS, and LoRa modules. An alert is transmitted to the train, where an LCD display and a buzzer notify the locomotive pilot to take appropriate action. The system offers an efficient solution to prevent collisions, thereby enhancing railway safety and reducing fatalities.

Keywords— Animal-train collision, Early warning system, Convolutional Neural Network, Buzzer alert, Railway safety, Accident prevention, Railway Safety, Alert system.

I. INTRODUCTION

Animal-train collisions pose a serious challenge in areas where railway tracks intersect wildlife habitats. These incidents often result in significant harm to wildlife, particularly endangered species, leading to injuries, fatalities, and the death of animals [1-2]. From an operational perspective, these collisions cause disruptions to railway services, infrastructure damage, and financial losses. The increasing prevalence of such accidents calls for urgent and effective solutions to protect wildlife and ensure railway safety.

One of the primary contributors to these collisions is the presence of railway tracks in high-risk zones known as hotspot, where animal pathways cross the tracks. Poor visibility during nighttime, adverse weather conditions, and sharp track curves limit the reaction time available to train operators, making collisions almost unavoidable. Existing solutions, such as fencing, manual patrolling, or acoustic deterrents, are often expensive, inefficient, and impractical for large-scale implementation. This underscores the need for innovative and automated approaches to address the issue effectively.

Recent developments in artificial intelligence, particularly Convolutional Neural Networks (CNNs), offer a powerful solution to this problem. CNNs are highly effective in object detection and classification, making them ideal for identifying animals near railway tracks. By utilizing CNN technology, an automated system can be developed to detect obstacles in real

time and provide timely alerts to train operators, significantly reducing the likelihood of collisions.[3-6]

This paper introduces a novel concept for an animal-train collision avoidance system that integrates CNN-based object detection with a hardware setup. The system operates by first using an IR sensor to detect obstacles along the railway tracks. Once an obstacle is identified, a camera is activated to capture an image, which is then analyzed using a pre-trained CNN model. If the detected object matches a predefined category, such as an animal or human, the system triggers alerts. Notifications are sent to the train station via LoRa system [7-9], warning messages are displayed on an LCD screen, and remote alerts are sent through GSM.

By combining CNN image processing with efficient hardware integration, the proposed system provides an effective solution to mitigate animal-train collisions. This approach not only enhances railway safety but also plays a crucial role in wildlife conservation by preventing accidents in critical hotspot. This paper details the design, methodology, and potential impact of this innovative solution, emphasizing its significance in wildlife protection and railway operations.

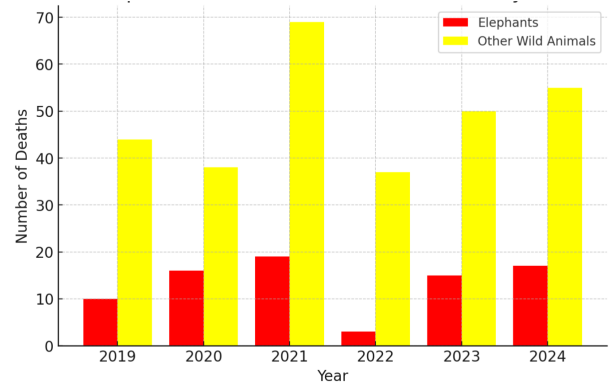


Fig. 1. Deaths of elephants & other wild animals on railway tracks in India

Figure 1 illustrates the number of deaths of elephants and other wild animals on railway tracks in India from 2019 to 2024.

- General Trend:

The number of deaths fluctuates over the years for both elephants and other wild animals. Wild animals consistently have a higher number of deaths compared to elephants.

- Elephant Deaths:

The deaths of elephants remain relatively lower than other wild animals. The highest number of elephant deaths is observed in 2021, followed by a decline in 2022. The deaths increased again in 2023 and 2024 but remain lower than in 2021.

- Deaths of Other Wild Animals:

The numbers are significantly higher than those of elephants. A sharp peak is observed in 2021, where the deaths of wild animals reach nearly 70. The lowest count for wild animal deaths is in 2022, but it rises again in 2023 and 2024.

II. LITERATURE SURVEY

Ranjith.A in paper [1] employs deep learning neural networks to detect animals on railway tracks. Their system integrates DNNs with IoT sensors such as cameras, ultrasonic, thermal, and infrared sensors to identify and track animals in real time. Using cloud-based AI processing and wireless networks, alerts are sent to train operators, helping mitigate collisions and enhancing railway safety, particularly in wildlife-prone areas.

Sabnis O.V & L.R in [2] focus on railway safety at unmanned railway crossings, proposing an object detection system that identifies potential obstacles such as humans, animals, or vehicles. By utilizing machine learning-based object detection and image processing techniques, the system ensures real-time monitoring and warning mechanisms. The study highlights the importance of edge computing in achieving faster and more reliable safety interventions at railway crossings.

Carvalho, F., et al in [3] review different strategies to monitor and mitigate wildlife mortality on railway tracks. They discuss various mitigation techniques, including fencing, wildlife overpasses, underpasses, and warning systems, to prevent accidents involving animals. By analyzing wildlife behavior and railway infrastructure interactions, the study offers recommendations for designing railway systems that minimize ecological impact.

Malo et al. in [4] explore the unintended impact of railway infrastructure on bird mortality, specifically how railway poles act as pitfall traps for birds. Their study investigates the design flaws in railway poles that lead to bird entrapment and mortality. They suggest modifications to railway infrastructure to make it more bird-friendly and prevent unnecessary wildlife casualties.

Lin et al. in [5] introduce the Focal Loss function, a technique to improve dense object detection in scenarios with severe class imbalance. Traditional object detection models struggle to recognize small objects within large backgrounds, which is particularly challenging in applications like autonomous driving, surveillance, and medical imaging. The study modifies existing loss functions in models such as Faster

R-CNN and SSD, improving detection accuracy, especially for hard-to-detect objects.

N. Mamat, M.F Othman & Yakub in paper [6] address the critical issue of safeguarding agricultural areas from animal intrusions, which can lead to significant crop damage and economic losses. The researchers propose utilizing the You Only Look Once version 5 (YOLOv5) deep learning model to detect and identify four specific categories of animals commonly responsible for such intrusions. YOLOv5 is renowned for its balance between speed and accuracy in object detection tasks. The study demonstrates that this approach can effectively generate high-precision detection results, thereby offering a viable solution for real-time monitoring and protection of farming areas against animal-related disruptions.

Noreen, Umber, Bounceur, Ahcène, and Clavier, Laurent, in paper [7], investigates LoRa, a Low Power Wide Area Network (LPWAN) protocol designed for Internet of Things (IoT) applications. LPWANs are pivotal in enabling large-scale wireless sensor networks, offering cost effectiveness, extensive coverage, and energy efficiency, making them ideal for smart city implementations. LoRa utilizes a spread spectrum technique that employs the entire channel bandwidth for signal transmission, enhancing resistance to channel noise, frequency shifts, Doppler effects, and fading. The study delves into the influence of three fundamental LoRa parameters Code Rate (CR), Spreading Factor (SF), and Bandwidth (BW) on data rate and time on air, providing an in depth analysis of how these parameters affect the performance and efficiency of LoRa networks.

Saari et al., in paper [8], presents a comprehensive literature review of LoRaWAN technology, focusing on its applicability in the Internet of Things (IoT) and sensor networks. The authors systematically analyzed over fifty research papers to categorize existing studies and identify application trends within LoRa research. They observed that recent research predominantly centers on performance evaluations and real-world deployments through prototype implementations and experiments. The study also provides recommendations for researchers and practitioners on leveraging LoRa-based technologies effectively in developing IoT systems and solutions.

S. Devalal and A. Karthikeyan in paper [9] by provides a comprehensive examination of LoRa (Long Range) technology, a prominent Low Power Wide Area Network (LPWAN) solution designed for Internet of Things (IoT) applications. LoRa enables long-range communication by utilizing chirp spread spectrum modulation, allowing devices to transmit data over distances ranging from a few kilometers in urban settings to up to 15-30 kilometers in rural areas. This technology is particularly advantageous for applications requiring low data rates and extended battery life, as it supports low-power, long-range, and secure data transmission. The authors delve into the technical aspects of LoRa, including its modulation schemes and network architecture, and discuss its potential applications across various industries, such as smart cities, environmental monitoring, and agriculture. They also address the challenges associated with deploying LoRa networks, such as interference

management and scalability, and suggest potential solutions to enhance network performance.

Kellenberger, Volpi, and Tuia in paper [10], address the challenge of efficiently detecting large animals in aerial imagery to combat illegal wildlife poaching. They developed a specialized object detection method utilizing Convolutional Neural Networks (CNNs) tailored for Unmanned Aerial Vehicle (UAV) images. Tested on data from Namibia's Kuzikus wildlife reserve, their approach significantly outperformed existing models in precision and achieved a processing speed exceeding 72 images per second, enabling real-time application in wildlife monitoring and anti-poaching efforts.

TABLE I
STUDIES DONE FOR THE PROPOSED SYSTEM

Authors	Title	Contributions
Ranjith, A., Vijayaragavan, S. P., & Muthukumaran, N.(2022)	An IoT-Based Monitoring System to Detect Animals on Railway Tracks Using Deep Learning Neural Networks	Developed an IoT-based system using deep learning to detect animals on railway tracks, reducing collision risks.
Sabnis, O. V., & L.R. (2019)	A Novel Object Detection System for Improving Safety at Unmanned Railway Crossings	Proposed an object detection system for railway crossings to enhance safety using machine learning techniques.
Gómez, A., Salazar, A., & Vargas, F. & N. Kumar (2017)	Towards Automatic Wild Animal Monitoring: Identification of Animal Species in Camera-Trap Images Using Very Deep CNNs	Applied deep CNNs to automate species identification in camera-trap images, improving wildlife monitoring efficiency.
Mamat, N., Othman, M. F., & Yakub, F. (2022)	Animal Intrusion Detection in Farming Area using YOLOv5 Approach	Used YOLOv5 for real-time animal intrusion detection in agricultural areas, enhancing farm security.
Ghodhbane, R. M., Hernandez, A., & Janfaoui, S. (2024)	A Pressure and Temperature Wireless Sensing Network Communicating with LoRa Modulation	Designed a LoRa-based wireless sensing network for remote monitoring of pressure and temperature.
Malayao, J. G., Bungay, I. M., & Yumang, A. N.(2024)	Image Classification of Gamefowl Breeds Using MobileNet V2	Used MobileNet V2 for accurate classification of gamefowl breeds, optimizing lightweight deep learning models. .
Kellenberger, B., Volpi, M., & Tuia, D.(2017)	Fast Animal Detection in UAV Images Using Convolutional Neural Networks	Developed a CNN-based method for rapid detection of animals in UAV images, aiding wildlife conservation.

III. METHODOLOGY

A. HARDWARE DEVELOPMENT

The hardware development of the animal-train collision avoidance system involves creating a physical setup capable of detecting obstacles and communicating alerts in real time. The design integrates multiple components, each serving a specific

role to ensure smooth operation. The block diagram in Figure 2 shows components.

The IR sensor acts as the primary detection unit, continuously monitoring for the presence of obstacles. When an object is detected within its range, it sends a signal to activate the camera module. The camera module is responsible for capturing high-resolution images of the detected object. This image is then analyzed to determine whether the object is an animal, human, or another obstacle.

The NodeMCU microcontroller is at the core of the system, serving as the central processing unit. It receives signals from the IR sensor and controls the camera's operation. It also processes the output from the CNN-based detection system to decide whether to generate an alert. The LCD display, connected to the Arduino Nano, at the receiver unit, provides visual alerts by displaying messages such as "Obstacle Detected," which can be seen by nearby personnel.

For communication, LoRa module and GSM module is used to transmit alerts wirelessly to the train operator or control station. This ensures that notifications are sent quickly, allowing train operators to take preventive actions. The system sends notifications to mobile devices with location of the detected area.

The entire system is powered by a stable DC power supply, ensuring uninterrupted functionality. The hardware components are assembled and tested for seamless integration. The designed system ensures high reliability, efficiency, and real-time performance, making it a practical solution for animal-train collision avoidance.

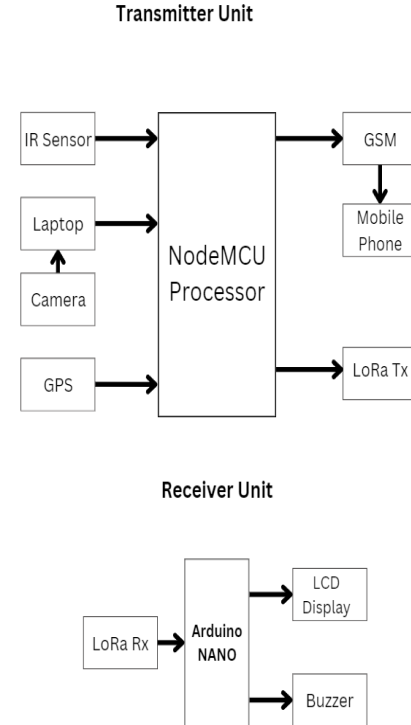


Fig. 2. Block diagram of the System

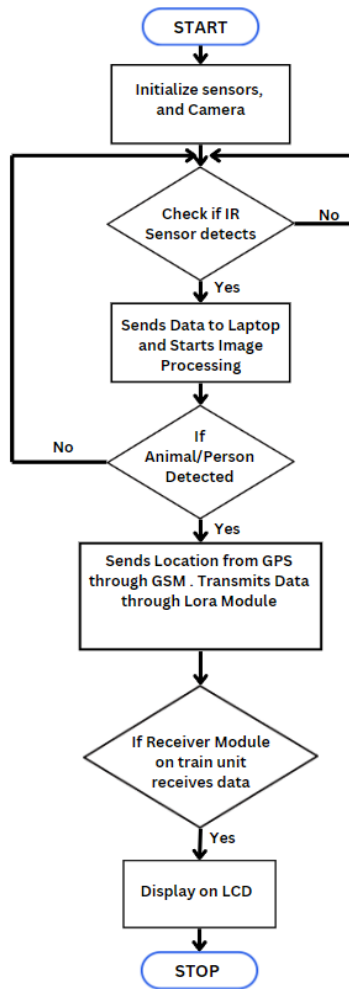


Fig. 3. Flow diagram of the system

The flow diagram of involves many steps from detection of obstacle by IR sensor to image processing and alert mechanism. The process is shown in Figure 3.

System Initialization

When the system is turned ON, all sensors are initialized.

- IR sensor activates and waits for obstacle to trigger it.
- GPS module connects to the satellite, acquiring the latitude and longitude of the installed area.
- GSM module with the SIM card connects to the nearest base station.
- LoRa module activates.
- The receiver unit, implemented in train, is always ON and waits for LoRa transmitted signal.

Processing

- The system constantly monitors if the IR sensor is triggered.
- If the IR sensor gets triggered by any obstacle, the system activates the implemented camera and starts video processing.

- The system implemented for CNN processing checks if the obstacle detected matches the pre-defined data set. If matched, it initialize further processing.

Alert Mechanism

- The system sends an SMS through the GSM module implemented in the system, which contains the number of the railway station authority (station master). The LoRa module sends signal at the same time.
- The station master receives a message that contain the system number along with the location of the system.
- Through this, the authorities can know where the animals are located.

Train Receiver Unit Processing

- As the train enters the LoRa transmitter range, the LoRa receiver detects the signal.
- The Arduino Nano present in the receiver processes the signal and displays "Obstacle Detected" in the LCD display.
- The buzzer implemented makes noise so that the loco pilot gets warning and notices the display.

B. SOFTWARE DEVELOPMENT

The software development of the system is centered around implementing a robust Convolutional Neural Network (CNN) for object detection and establishing communication protocols for real-time alerts. The CNN is trained with a dataset containing images of animals, including elephants, cows, dogs, goats, bears, and humans. This training enables the system to accurately classify objects captured by the camera module, ensuring reliable detection even in challenging environments such as low-light conditions, poor weather, or obstructed views due to vegetation or terrain.

The software begins by initializing the hardware components, such as the IR sensor and camera. When the IR sensor detects an obstacle, it signals the software to activate the camera. The captured image is processed using the pre-trained CNN model, which analyzes the image to identify features that match the trained categories. If the object is classified as a predefined obstacle, such as an animal or human, the software triggers the alert mechanism.

The alert mechanism involves multiple steps. First, the software sends a signal to the LoRa module to transmit a warning message to the train operator or station. Simultaneously, the Arduino Nano displays a visual alert on the LCD screen with the message "Obstacle Detected." A buzzer is also activated to ensure the alert is noticeable. Additionally, the system sends an SMS notification to railway authorities through the GSM module, which includes the GPS coordinates of the detected object's location.

The software is developed using Python for CNN implementation and image processing, utilizing libraries like TensorFlow for model development. The microcontroller's programming is done in embedded C, ensuring compatibility with the Arduino Nano. Extensive testing is conducted to validate the system's accuracy, response time, and communication reliability.

Performance metrics such as detection speed, false positive rate, and system robustness are analyzed to ensure real-time functionality, making it a vital component of the collision avoidance solution.

IV. RESULT AND ANALYSIS

The system represented in Figure 4 is an innovative and practical approach for preventing animal-train collisions by integrating image processing with wireless communication technologies. The system contains LoRa transmitters and receivers for efficient long-range data transmission, over 7 kilometers, ensuring that alerts are sent in real time. The GPS module provides location tracking.

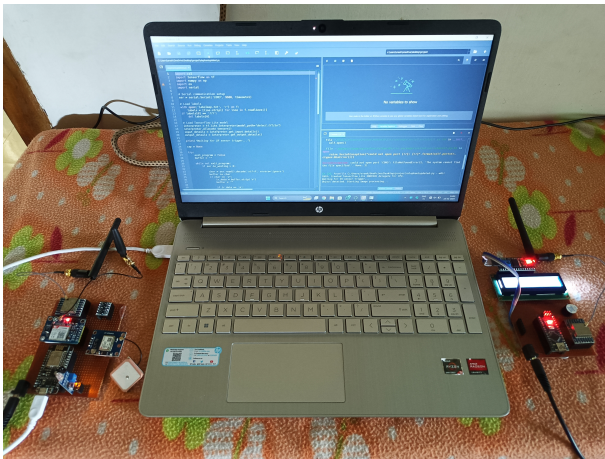


Fig. 4. System

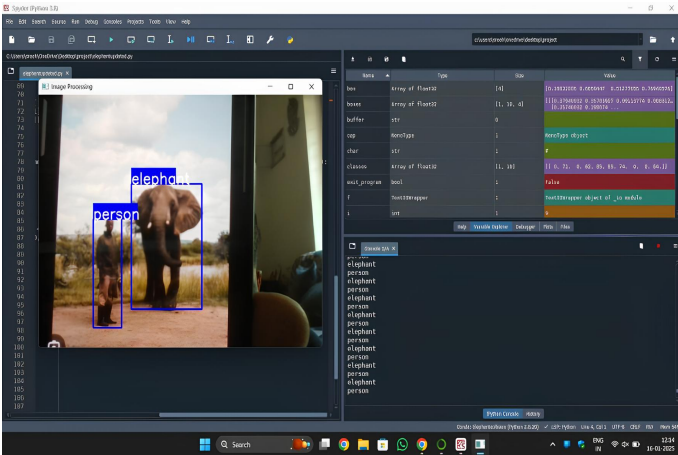


Fig. 5. CNN detecting an elephant and a person together

- Figure 5 shows that the Spyder IDE detects and classifies objects in an image displayed in a separate window.
- The detected objects include an elephant and a person both outlined with bounding boxes and labelled.
- Figure 6 shows that the IDE processing another image, this time recognizing a cow.

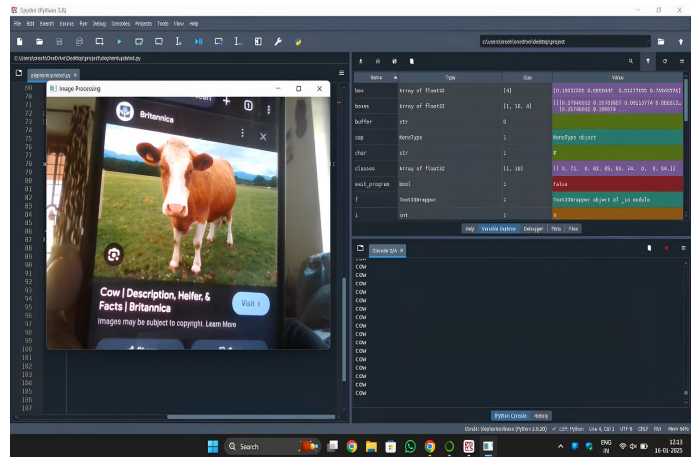


Fig. 6. CNN detecting a cow

- The detected object is labelled as "COW", repeated in the console output.

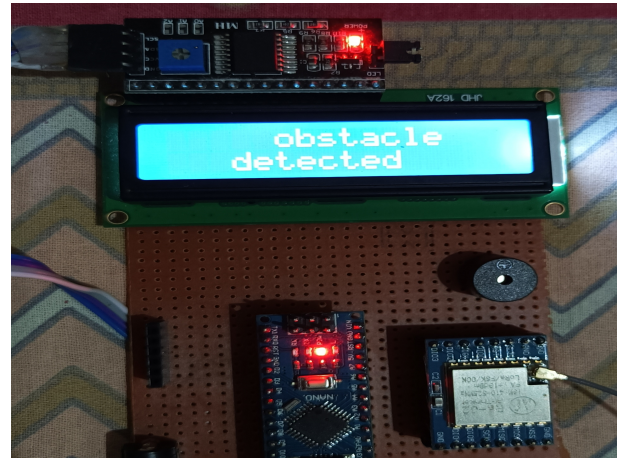


Fig. 7. LCD display showing alert message

- Figure 7 shows the message displayed on the LCD display which is received through the LoRa antenna. The buzzer also makes sound, notifying the loco-pilot.
- Figure 8 shows the received SMS through GSM module which include the location of the detected area.

The graph in figure 9, was generated to assess classification performance across different classes. The system achieved precision, recall, and scores of 92%, 90%, and 91% for elephants; 95%, 94%, and 94.5% for persons; 87%, 85%, and 86% for goats; 90%, 88%, and 89% for bears; 89%, 87%, and 88% for cows; and 93%, 92%, and 92.5% for dogs. The system performed best in detecting persons and dogs due to their distinct features. Minor misclassifications were observed between goats and cows, likely due to their similar body structures. Further dataset augmentation and fine-tuning could improve differentiation between visually similar animals.

The communication and alert system was also evaluated. LoRa transmission successfully sent alerts over a range of up

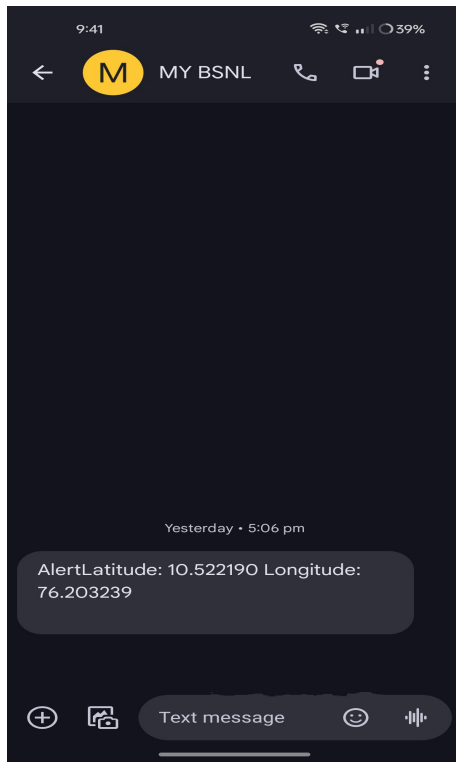


Fig. 8. Alert received through SMS with location of the detected place

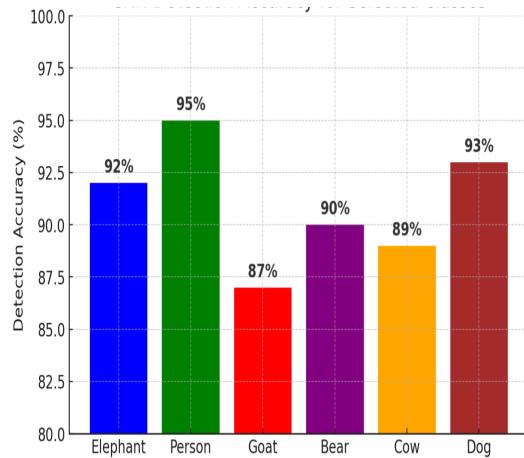


Fig. 9. CNN detection accuracy for selected classes

to 7 km. GSM alerts delivered SMS notifications with GPS coordinates in under 5 seconds. The train operator display showed real-time detections with an alert sound within 2 seconds of detection.

The overall system impact is significant in wildlife and railway safety. The early warning system can substantially reduce animal-train collisions, particularly for endangered species like elephants. Train operators receive timely alerts, allowing emergency braking to prevent accidents. The system is scalable and can be deployed across multiple railway tracks with minor modifications for regional adaptability.

The system demonstrates high accuracy in detecting animals and persons near railway tracks. It ensures reliable real-time alerts through LoRa and GSM, effectively preventing animal-train collisions. Future improvements could focus on enhanced night vision capabilities and adaptive learning for new animal species.

V. CONCLUSION

The warning system presented in this paper presents an innovative and practical approach to preventing wildlife-train collisions by integrating MobileNet V2 CNN based image processing with wireless communication technologies. The system employs LoRa transmitters and receivers for efficient long-range data transmission, ensuring that alerts are relayed in real time. Additionally, the GSM module provides location tracking, enabling timely intervention to mitigate potential accidents.

The CNN-based detection system is trained on a dataset comprising images of elephants, persons, goats, bears, cows, and dogs. The model underwent extensive training and validation to optimize detection accuracy. The final training accuracy was 95%, with a validation accuracy of 92%. The training loss settled at 0.2, while validation loss reached 0.3, indicating that the model successfully generalizes to unseen data, ensuring reliable real-time detection.

This device has significant social and environmental relevance. Animal-train collisions pose a serious threat to both biodiversity and railway operations, leading to loss of animal life, damage to railway infrastructure, and disruptions to transportation networks.

This solution aligns with global efforts toward sustainable development by CNN technology to address real-world challenges. With further refinement and scalability, it can be adapted to various railway networks, contributing to a smarter, safer, and more environmentally responsible transportation system.

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